

Overload Alarm: Pareto Optimization of the Economic-Social-Environmental Triangle

Summary

The rapid growth of tourism has led many countries and regions to recognize the tension between environmental preservation and the short-term economic benefits of traditional tourism models. The overload alarm of tourism industry prompts the question: How should tourism evolve in the future? **Sustainable tourism** has emerged as a crucial solution to this dilemma. This paper develops a **multi-objective planning model** that integrates economic, social, and environmental factors, tailored to the current state of the tourism industry in Juneau and its development goals.

In Sub-model I, **economic benefit** is the primary optimization objective. Considering the diminishing marginal returns and increasing marginal costs associated with a rising number of tourists, we developed a mathematical relationship between economic benefits and costs as a function of tourist volume. Through a **cost-benefit analysis (CBA) model**, we then defined the feasible domain of economic constraints and determined the maximum economic benefit.

In Sub-model II, **social benefit** is the second optimization objective. It is assessed through social satisfaction analysis and monetization. Social satisfaction is influenced by residents' living pressure and regional congestion. We use the potential benefit per unit of social satisfaction as the conversion factor and model social benefit changes with tourist numbers. A **social welfare optimization model** identifies the feasible domain and calculates the maximum social benefit.

In Sub-model III, **environmental benefit** is the third optimization objective. Focusing on glacier degradation, the primary environmental issue in Juneau. Given the link between glacier degradation and carbon emissions, the carbon footprint is chosen as the key environmental indicator. The monetization of this effect follows a similar method to that of social satisfaction. Through **ecological carrying capacity analysis**, we determine the feasible range for environmental benefits and calculate the maximum achievable value.

Furthermore, we integrate the three preceding models and account for the coupling relationships among them. Using the **SLSQP method**, we determine the global **Pareto-optimal solution** for tourist numbers and overall benefits. Specifically, when the number of tourists reaches **1.32 million**, the economic measure of the municipal weighted total benefit is **10 million USD**, while the net income for Juneau amounts to **576 million USD**.

Finally, we analyze the sensitivity of key model parameters, as well as the sources of additional government revenue and spending programs. We also write a memo to the Tourist Council of Juneau and extend the model to Jiuzhaigou, China. The results show that our model has good usability and robustness.

Keywords: Sustainable Tourism, Management Measures, Multi-objective Planning Model, Pareto Optimization, SLSQP